

Student: Jonatan Rabjerg
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Instructor: Andreas Basse-O'Connor
Course: Calculus Beta

Assignment: MyLab Math - Uge 13

Find the derivative of the function at P_0 in the direction of $\mathbf{A}$.

$$f(x,y) = 2xy - 2y^2, \quad P_0(5,5), \quad \mathbf{A} = 5\mathbf{i} - 3\mathbf{j}$$

The derivative of a function at any given point in the direction of a vector $\mathbf{A}$ is equal to the dot product of $\mathbf{u}$, the unit vector in the direction of $\mathbf{A}$, and the gradient of the function at that point.

Start by finding the gradient of the function. The gradient is given by, $\nabla f = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j}$.

$$\frac{\partial f}{\partial x} = 2y$$

$$\frac{\partial f}{\partial y} = 2x - 4y$$

$$\nabla f = 2y\mathbf{i} + (2x - 4y)\mathbf{j}$$

Evaluate the gradient at the given point, $P_0(5,5)$.

$$\nabla f = 2y\mathbf{i} + (2x - 4y)\mathbf{j}$$

$$\nabla f = 2(5)\mathbf{i} + (2(5) - 4(5))\mathbf{j}$$

$$\nabla f = 10\mathbf{i} - 10\mathbf{j}$$

Now, find the unit vector $\mathbf{u} = \frac{\mathbf{A}}{|\mathbf{A}|}$. Find the magnitude of $\mathbf{A}$.

$$|\mathbf{A}| = \sqrt{5^2 + 3^2}$$

$$= \sqrt{34}$$

$$\text{Since } |\mathbf{A}| = \sqrt{34}, \quad \mathbf{u} = \frac{5\mathbf{i} - 3\mathbf{j}}{\sqrt{34}}$$

Use the dot product and the gradient to find the derivative of $f(x,y)$ in the direction of $\frac{5\mathbf{i} - 3\mathbf{j}}{\sqrt{34}}$. In this case, the dot product is given by the following expression.

$$D_{\mathbf{u}}f = \frac{\partial f}{\partial x}u_x + \frac{\partial f}{\partial y}u_y$$

$$D_{\mathbf{u}}f = \frac{(10)(5) + (-10)(-3)}{\sqrt{34}}$$

$$D_{\mathbf{u}}f = \frac{40\sqrt{34}}{17}$$

Thus, the derivative of the function, $2xy - 2y^2$ at $P_0(5,5)$ in the direction of $5\mathbf{i} - 3\mathbf{j}$ is $\frac{40\sqrt{34}}{17}$.